Chapter 2, Problem 4RQ

Problem

The maximum current that a 2W, 80 kΩ resistor can safely conduct is:

(a) 160 kA

(b) 40 kA

(c) 5 mA

(d) 25 μA

Step-by-step solution

Step 1 of 3

Write the expression for power dissipated by a resistor R.

$$p = i^2 R$$

$$i^2 = \frac{p}{R}$$

$$i = \sqrt{\frac{p}{R}} \quad \text{...... (1)}$$

Step 2 of 3

Determine the maximum value of current.

Substitute $2\ \text{W}$ for p and $80\ \text{k}\Omega$ for R in equation (1).

$$\begin{aligned} i &= \sqrt{\frac{2}{80\times 10^3}} \\ &= \sqrt{25\times 10^{-6}} \\ &= 5\times 10^{-3} \\ &= 5\ \text{mA} \end{aligned}$$

The current value $(5\ \text{mA})$ does not match with option-(a), option-(b) and option-(d).

Hence, the options (a), (b) and (d) are wrong.

Step 3 of 3

From the calculations, the maximum current that a $2\ \text{W},\ 80\ \text{k}\Omega$ resistor can safely conduct is, $\boxed{5\ \text{mA}}$.

Hence, the correct option is $\boxed{(\text{c})}$.